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- Subject: ML.

# * Assignment No: 07 *

Q-1]

=> Reason 1: Gradient based second order optimization built in regularization

- XGBoost fits trees to gradients and Hessians which gives more accurate step directions for a wide range of losses
- It also exposes L1/L2 regularization on leaf weights learning-rate, max depth and min-child-weight all of which reduce overfitting

Reason 2: Engineering and algorithmic improvement for real data

- sparse aware split finding missing value handling column/row subsampling and an efficient approximate split algorithm make training robust & fast on messy
- parallelization and cache friendly data layout let XGBoost train larger models quickly & reliably in practice

Trade off introduced:

- Complexity and compute :- more hyperparameters to have & higher training cost vs simpler learners
- Interpretability : more opaque than single trees or linear models.
- Risk of miss tuning

Q2]

=> why majority vote can make thing worse

- Correlated errors : If the three models make similar mistakes.
- Bad calibration or bias - If one model is systematically biased or miscalibrated it can disturb the vote
- Overfitting /data leakage - Naively training the ensemble on the same training set without proper out of fold stacking leaks information

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* Better ways to combine :

i] Stacking with out of fold predictions
ii] Soft voting / weighted averaging of probabilities
iii] Blending / cross - validating ensemble
iv] Increase diversity
v] Model selection + regularization at meta level.

* practice checklist to fix your degraded ensemble.

- check based - model correlation on validation set
- Switch to soft voting with probability calibration
- Implement 5 - fold OOF stacking with a simple meta model (logistic / reg)
- validate ensemble on a holdout set and tune ensemble weights / meta regularization

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